Zac Cross

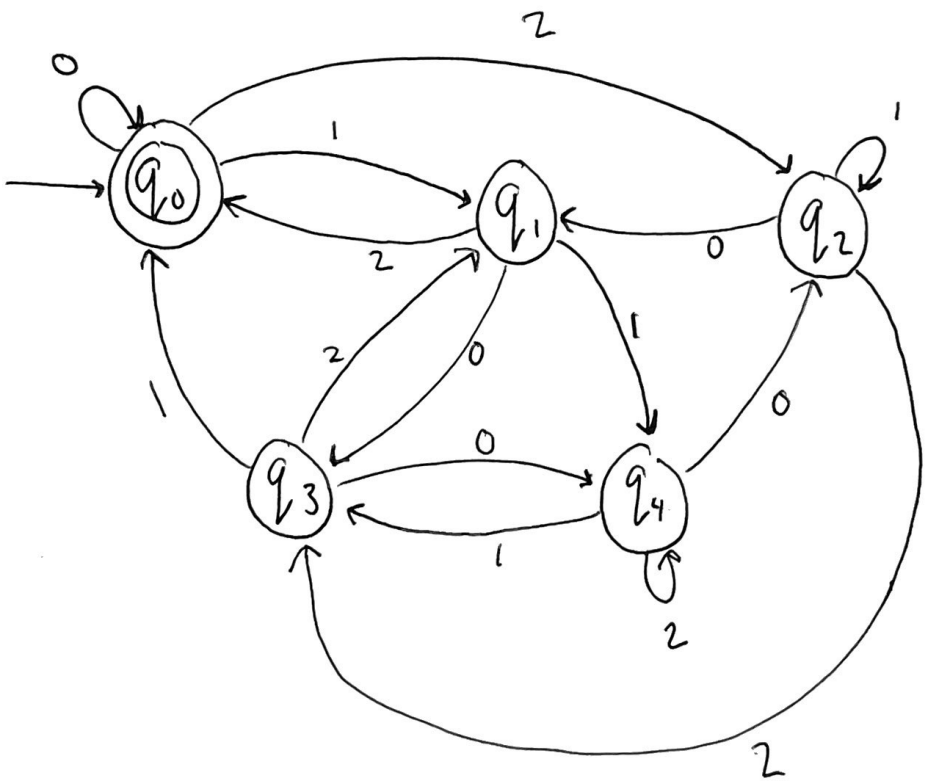
CS 520

19 Oct 2025

HW 2

1.) $\{w \in \{0,1,2\}^* \mid w$ represents an integer in base 3 divisible by $5\}$

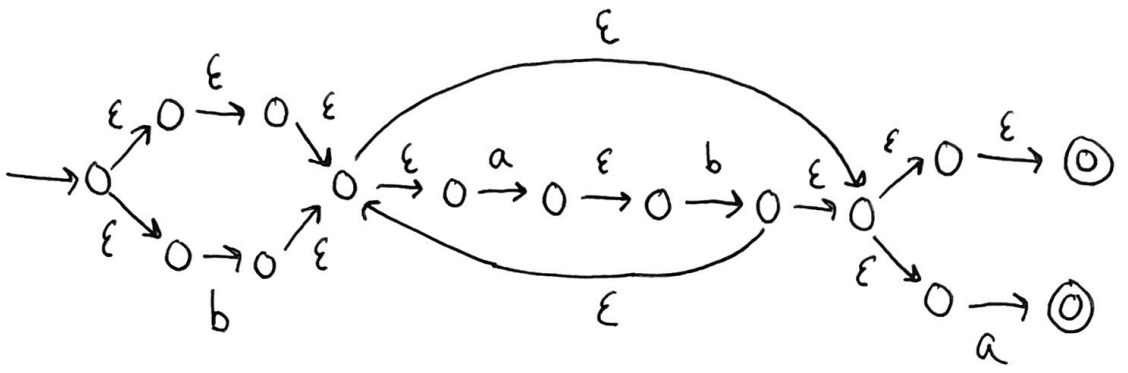
DFA

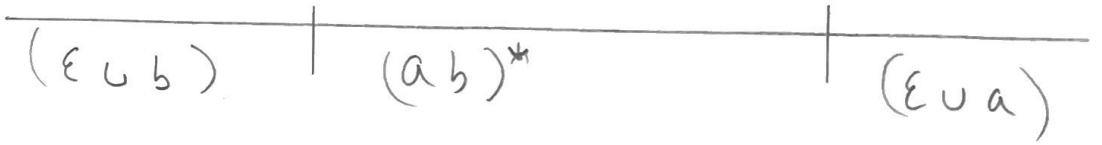


2.)

a.) Reg Ex: $(\varepsilon \cup b)(ab)^*(\varepsilon \cup a)$

<u>NFA</u>

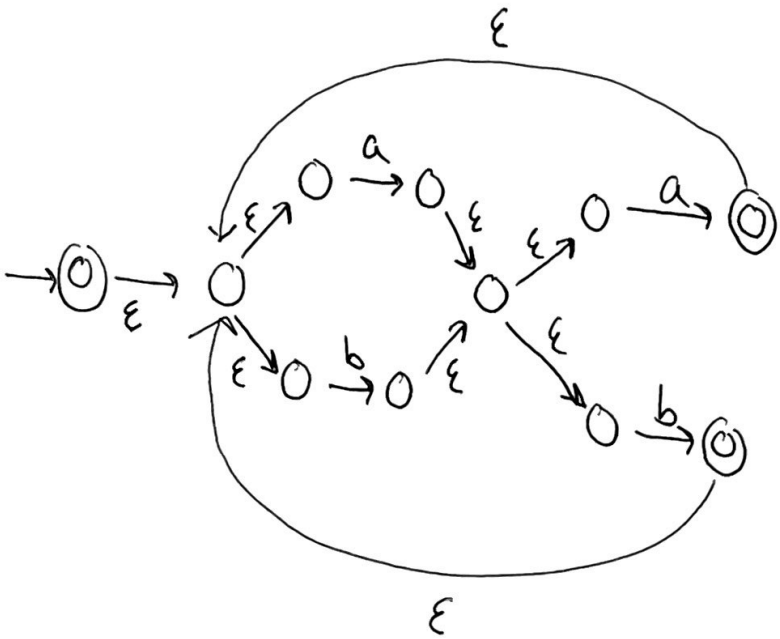




2.)

b.) <u>Reg Ex:</u> $((a \cup b)(a \cup b))^*$

<u>NFA</u>

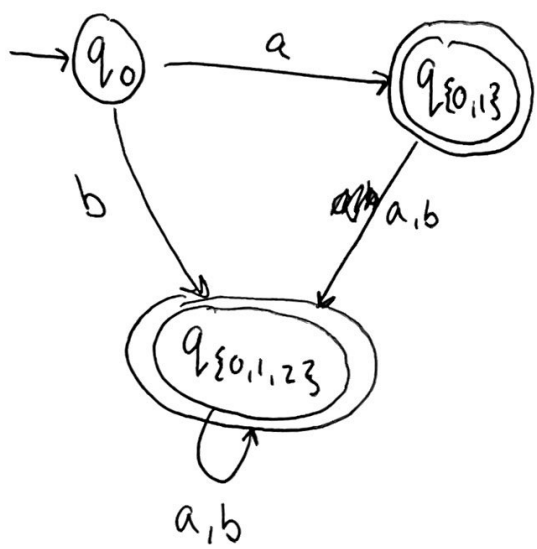


3.)

Helper Table:

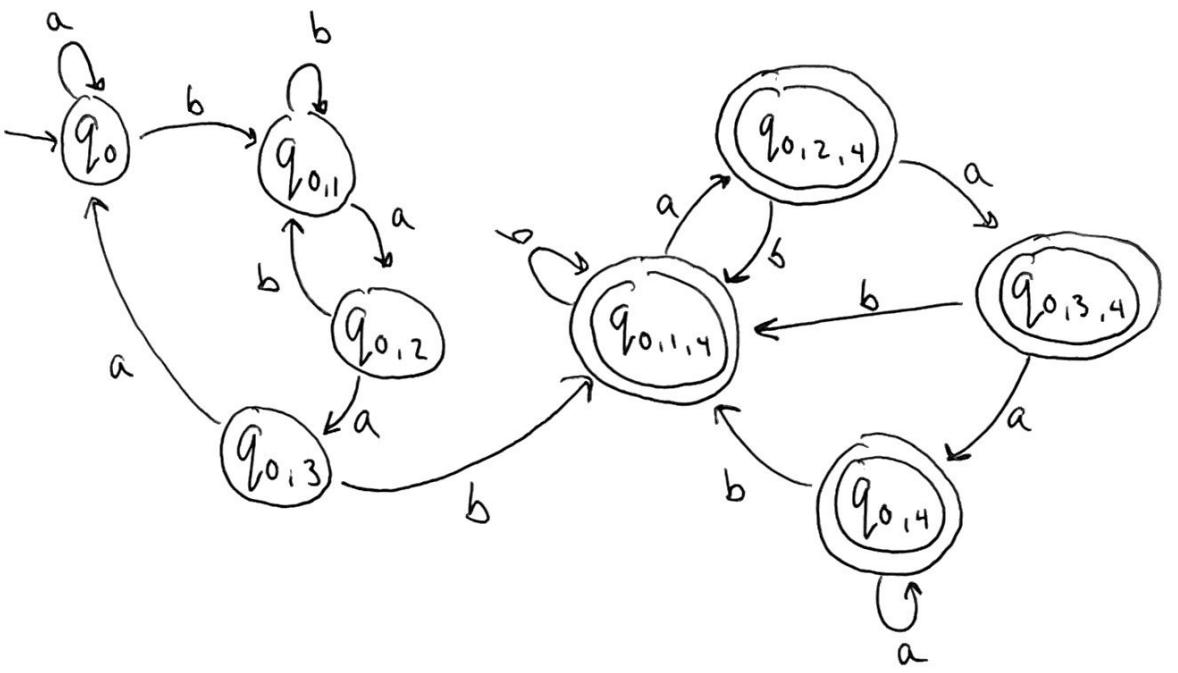
	a	b
q_0	0,1	0,1,2
q_1	0,1,2	0,1
q_2	$\emptyset$	$\emptyset$

DFA



4.)

Helper Table:

	a	b
0	0	0,1
1	2	∅
2	3	∅
3	∅	4
4	4	4

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